# Ellipse Polar Encoding for Oriented SAR Ship Detection

Jun Liu, Lin Liu, and Jiarong Xiao

***Abstract*—Ship detection in synthetic aperture radar (SAR) images using deep neural networks often relies on horizontal bounding box, which fail to capture ship orientation and aspect ratio accurately. To address this limitation, oriented ship detection methods based on oriented bounding box (OBB) are gaining attention. However, most available existing OBB methods suffer from boundary discontinuity problem, leading to convergence problems and unsatisfied orientation detection performance. In this article, we propose an innovative oriented SAR ship detection method using ellipse polar encoding (EPE). By representing the ship detection box as an ellipse and employing a set of vectors from the center to the boundary as encoded parameters, our method exhibits smooth variations, enhances convergence, and efficiently decodes into OBB. We further develop a lightweight and effective oriented SAR ship detection network based on this methodology. To account for SAR image characteristics, such as speckle and deformation causing deviations from true ellipses, we introduce an intersection over union weighted EPE loss. The experimental results on the rotated ship detection dataset in SAR images demonstrate the effectiveness of our proposed method in significantly improving detection performance compared with other oriented target detection methods.**

***Index Terms*—Oriented bounding box (OBB), polar encoding (PE), rotated ship detection dataset in SAR (RSDD-SAR), ship detection, synthetic aperture radar (SAR).**

## I. Introduction

SYNTHETIC aperture radar (SAR) is a highly effective space-based Earth observation technology widely employed in remote sensing for ground and sea surface targets. It provides all-day, all-weather imaging capabilities, a vast observation area, and high imaging resolution. SAR images are celebrated for their resilience compared with natural optical images, rendering them indispensable for various applications. In practice, ship detection-based SAR images play a crucial role in some specific fields, including marine target monitoring, maritime rescue, port security, etc. [1], [2], [3], [4], [5].

Ship detection-based deep learning in SAR images is essential for SAR data intelligence interpretation. Ships appearing in SAR images typically exhibit elongated shapes with high aspect ratios and can be oriented in arbitrary directions. Current common SAR ship detection algorithms often utilize a horizontal bounding box (HBB) to represent the ship's position and size [1], [2], [3]. Although HBBs can effectively enclose ship regions, they fail to capture the diverse poses and shapes that ships can adopt. Furthermore, due to SAR imaging characteristics, coastal buildings may exhibit strong scattering points that resemble ship features, which adds complexity to detecting inshore ships.

To address this challenge, researchers have recently turned their attention to oriented SAR ship detection algorithms employing oriented bounding box (OBB) [6], [7], [8]. Unlike HBB, OBB is a rectangular box that incorporates a rotation angle, enabling it to align with the main axis of the ship. By utilizing OBB, redundant information is minimized, and overlapping with other boxes is avoided, generating a more accurate representation of the ship's length, width, and orientation. Indeed, OBB-based ship detection methods overcome the limitations of HBB-based methods successfully.

In the literature review, OBB-based ship detection methods can be categorized into three primary categories: anchor-based methods [9], [10], [11], [12], [13], [14], angle-based methods [15], [16], and encoding-based methods [17], [18], [19]. Unfortunately, these methods are susceptible to the boundary discontinuity problem to some extent, caused by the periodicity of angle (POA) and the exchangeability of edge (EOE). The boundary discontinuity problem introduces a mismatch between predicted and ground truth instances during the training phase, leading to a degradation in performance. To mitigate this problem, researchers propose various methods [17], [18], [19]. However, these methods fail to provide a fundamental solution to the problem but partial alleviation. Furthermore, the adoption of these methods often results in complex model structures and decreased convergence performance.

To address the boundary discontinuity problem effectively, this article introduces the ellipse polar encoding (EPE) method and presents an end-to-end lightweight-oriented SAR ship detection network based on this method. In this method, the ship detection box is treated as an ellipse and encoded as a collection of vectors originating from the center and extending to the boundary. These length vectors, predicted by the network as encoded parameters of EPE, exhibit continuity and smoothness. The inherent continuity and smoothness of the encoded parameters aid in avoiding ambiguity during the training process and facilitate the convergence of the network. Furthermore, the encoded parameters can be efficiently decoded into OBB.

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Considering the specific characteristics of ships in SAR images, such as speckle and deformation, which may deviate from perfect ellipses, we propose utilizing a weighted encoded parameters loss function based on the intersect over union (IOU) metric. To evaluate the effectiveness of our method, we conduct experiments on the rotated ship detection dataset in SAR images (RSDD-SAR) [20]. Comparative analysis with other oriented target detection methods demonstrates the superior performance of our proposed method, achieving higher detection accuracy with a reduced number of model parameters. In summary, this article presents the following contributions.

1) We introduce an end-to-end lightweight-oriented SAR ship detection network that is specifically designed for efficient oriented SAR ship detection.
2) We propose the EPE method, which effectively tackles the boundary discontinuity problem in orientation detection. The EPE method ensures smooth convergence during training and enables fast decoding of the encoded parameters.
3) To enhance network convergence, we develop a loss function that incorporates an IOU weighting scheme for the encoded parameters.
4) To validate the effectiveness of our proposed approach, we conduct experiments on the publicly available dataset RSDD-SAR, which serves as a benchmark for evaluating oriented ship detection algorithms in SAR images.

## II. Related Work

### A. Target Detection Methods Based on HBB

HBB offers clear geometric properties and is easily parameterized, making it suitable for detecting targets with uniform dimensions. Target detection methods based on HBB can be categorized into two types: one-stage and two-stage. Two-stage algorithms, such as R-CNN [21] and Faster R-CNN [22], typically outperform one-stage methods due to their complex pipelines. Two-stage algorithms involve region proposal generation and subsequent object classification and localization, enabling the capture of finer target details. Lin et al. [23] propose that the primary reason for the low accuracy of one-stage algorithms is the issue of sample imbalance. Thus, they introduced the focal loss to enhance the performance. Subsequently, one-stage algorithms, such as YOLOv3 [24] and EfficientDet [25], significantly improved both accuracy and speed. Anchor-free detection methods further reduce model complexity and enhance speed. Tian et al. [26] introduce FCOS, which abandons anchors and introduces a multiscale matching strategy and a centerness branch, effectively enhancing detection accuracy. In addition, key point-based algorithms, such as CenterNet [27] and CornerNet [28], demonstrate the simplicity and effectiveness of anchor-free approaches.

### B. Boundary Discontinuity Problem

The boundary discontinuity problem emerges in oriented object detection methods based on OBB due to two key factors: POA and EOE. POA refers to the fact that when the angle of OBB approaches the upper or lower boundary of its range, the represented orientation becomes very similar, leading to angle discontinuity. EOE refers to the sudden change in the width or height of OBB at the boundary, resulting in a discontinuity in width or height. The boundary discontinuity problem arising from POA and EOE can lead to erroneous parameter loss even when the predicted OBB closely resembles the ground truth [8].

### C. Target Detection Methods Based on OBB

Currently, there are three main categories of oriented object detection methods using OBB: anchor-based methods, angle-based methods, and encoding-based methods.

The anchor-based methods generate a rotated anchor based on a common horizontal anchor and predict the target's OBB by estimating the error between the predicted anchor and the ground truth. Various improved methods are proposed by researchers to enhance the performance of anchor-based oriented object detection algorithms. Xia et al. [9] propose the R-faster R-CNN algorithm as an improvement over faster R-CNN, specifically targeting the detection of oriented objects, which they achieve this by regressing from horizontal anchor to OBB in the second stage of the detection pipeline. Ding et al. [10] further enhance the region proposal network (RPN) by converting horizontal regions of interest (ROI) into rotated ROI, enabling the extraction of features from objects with arbitrary orientations. Xie et al. [11] introduce the oriented RPN and oriented R-CNN head, incorporating orientation information into both the region proposal and classification networks, which contribute to improve detection speed. Han et al. [12] propose a rotation-invariant ROI alignment method to address the challenge of accurate orientation prediction. Their approach extracts rotation-equivariant and rotation-invariant features to improve orientation estimation. In addition, Han et al. [13] tackle the inconsistency between classification scores and localization accuracy by employing a feature alignment module and an oriented detection module, further enhancing the detection performance. Furthermore, Yang et al. [14] introduce a technique that re-encoded the positional information of bounding boxes onto the corresponding feature points through pixelwise feature interpolation. This approach enables feature reconstruction and alignment, contributing to improve detection performance. Despite these advancements, anchor-based methods still face certain challenges. First, the process of generating anchors is time-consuming, as a majority of them end up being negative samples, which significantly limits the algorithm's speed. Second, the design of anchors is complex and plays a crucial role in determining the quality of detection performance.

The angle-based approach extends HBB-based target detection by adding a prediction branch for the rotation angle, directly predicting the width, height, and angle of OBB. Zhu et al. [15] propose a method that represents the rotation angle using a 2-D periodic vector. They introduce a length-independent IOU metric and a periodic loss function to mitigate the negative impact of POA during network training. However, this method still requires manual parameter settings to address the boundary discontinuity problem. From a distinct perspective, Yang and Yan [16] propose a classification-based method for predicting

the rotation angle. They employ a periodic loss function to alleviate the boundary discontinuity problem associated with OBB regression. Nevertheless, this approach introduces a computational burden due to the classification task.

The encoding-based methods represent OBB using a group of parameters and the network predicts these parameters to decode and obtain the predicted OBB. Yi et al. [17] employ a representation of the OBB as a set of vectors from the center to the midpoints of its four edges. Xu et al. [18] take a two-step approach, first predicting the HBB and then estimating the distances between the vertices of the HBB and the vertices of the OBB. Although these methods address the principle of POA issue, they encounter discontinuity problems in the encoded parameters, particularly in cases involving alignment with the pixel grid. To tackle the boundary discontinuity problem, a fixed threshold is manually set to differentiate between HBB and OBB. In addition, an extra network branch is added to classify the bounding box as HBB or OBB, which causes complexity and reduces network convergence. Zhao et al. [19] adopt a regression approach on the polar coordinates of the OBBs four vertices for oriented detection. However, this method requires a large number of regression parameters and still experiences angle discontinuity at the boundaries. In addressing the spatial feature aliasing problem in oriented and densely arranged object detection, Guo et al. [29] propose the convex feature adaptation (CFA) method. This technique involves dynamic feature point sampling and iterative convex hull splitting.

### D. Ship Detection in SAR Images

In the field of SAR ship detection, both HBB-based detection methods and OBB-based detection methods are being extensively studied. Liu et al. [30] propose a method that integrates pyramid features extracted by a deep convolutional neural network (DCNN) into the sea–land segmentation and ship detection processes. Deng et al. [31] train a DCNN-based ship detector from scratch on SAR datasets, incorporating dense blocks and novel training losses. Gao et al. [32] improve the performance of the detection framework, CenterNet [27], by incorporating attention mechanisms and feature reuse strategies. Zhou et al. [33] employ a dual-pooling feature extraction module, a network structure compensating for data loss, and a novel loss function based on dual Euclidean distances, thereby enhancing SAR small ship detection accuracy. Bai et al. [34] propose a network that includes a spatial enhancement module, feature alignment module, and shallow feature reconstruction module to detect SAR ships in complex environments.

For OBB-based detection, An et al. [6] propose an improved OBB encoding scheme and a multilevel anchor setting to enhance the accuracy of position and orientation estimation. Pan et al. [7] utilize a rotated RPN to extract candidate target regions, followed by fine-tuning the OBB detection results using a multilevel cascaded network. Another approach by He et al. [8] employs the polar encoding (PE) method, where OBB is encoded as a group of polar samples at fixed intervals. By predicting the length values of the polar samples and decoding them, the predicted OBB is obtained. This method successfully addresses the boundary issue by utilizing a unique mapping between the OBB and the encoded parameters. However, it suffers from poor real-time performance due to the computational complexity of its decoding processes. Gao et al. [35] consider and leverage the geometric and scattering features of SAR ships. They transform the OBB into two distinct components: angle classification value for the ship's bow and a set of corresponding distance values using ellipse coding. Fortunately, this approach effectively resolves the boundary discontinuity problem. During training, these two prediction branches calculate their respective losses, respectively. During inference, they merge and decode the outputs of these two branches to generate the detection results, potentially introducing coupling errors.

## III. Proposed Method

This section introduces an oriented SAR ship detection method based on EPE. The architecture of the method is depicted in Fig. 1 and consists of three main components: the feature extraction network (Backbone), the feature fusion network (Neck), and the detection head (Head). The backbone processes the input image to extract features at various scales, which are then fused in the neck to produce fused feature maps. These fused feature maps are subsequently fed into the head, which outputs the center heatmap ($H$) [27] and the encoded parameters map of EPE ($E$). The resulting outputs are decoded to obtain the ship detection boxes.

To accurately represent SAR ship, the detection box is considered an equivalent ellipse. The method involves sampling a set of equiangular polar coordinate points along the ellipse's boundary and utilizing the lengths of these polar coordinates as the encoded parameters for regression training. During inference, the predicted encoded parameters are decoded to obtain boundary points that form an ellipse representing the ship detection box.

The training process involves encoding the ground truth information to generate the center heatmap and the encoded parameters, with corresponding loss computations for each prediction branch. During the inference stage, postprocessing is applied using nonmaximum suppression (NMS) [36] to refine the detection results. NMS eliminates redundant detections and ensures that only the most confident and nonoverlapping detections are retained.

This section will first present the detection network model, followed by a detailed explanation of the EPE and decoding methods. Next, we will explain the central heatmap label generate method and NMS. Finally, the loss functions used for network training will be described.

### A. Our Network Model

To evaluate the effectiveness and generalization capability of our proposed oriented SAR ship detection method, we select the widely used residual network (ResNet) [37] as the backbone and the feature pyramid network (FPN) [38] as the neck. In addition, we design a dual-branch detection head for predicting ship centers and the encoded parameters of EPE, as depicted in Fig. 1.

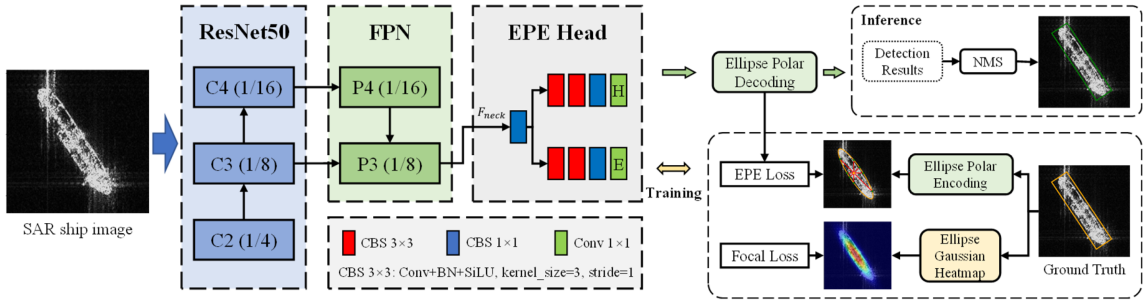



Fig. 1. Overall architecture of the proposed method.

ResNet [37] is a highly renowned network architecture. It introduces the concept of residual connections, facilitating the direct flow of information across layers and mitigating the network degradation problem. This architectural innovation enables the efficient training of deep networks, facilitating the extraction of intricate and informative features crucial for accurate target detection. Due to this characteristic, numerous object detection algorithms leverage ResNet as their backbone [8], [23], [26], [29]. Similarly, we believe that ResNet can effectively extract features from SAR images of ships, thereby facilitating accurate ship detection. In our method, we employ a modified ResNet50 backbone by removing the C5 stage. This decision is motivated by the simplicity and directness of our EPE method. By removing the C5 stage, the modified ResNet50 backbone still retains its strong feature extraction capabilities while reducing the overall model size and computational requirements. In Section IV, we will further conduct comparative experiments to validate the accuracy of our selection.

Due to the differences in the real physical sizes of ships and the resolution of SAR images, ships often exhibit multiscale characteristics in SAR images. To address this challenge and enhance adaptability to different ship scales, we employ the FPN [38] to fuse features extracted at various scales by the backbone. By integrating FPN into our approach, the detector effectively leverages both shallow features and deep semantic information, leading to improve detection performance. This fusion enables a comprehensive representation of the multiscale features present in SAR ship detection. Consequently, the output feature map resolution is reduced to one-eighth of the input image, providing a concise and informative representation for subsequent processing stages.

Based on the EPE method, we develop a dual-branch detection head to facilitate the prediction of ship centers and the encoded parameters. As illustrated in Fig. 1, the features ($F_{\text{neck}}$) obtained from FPN undergo initial processing through a CBS (Conv+BN+SiLU) module, culminating in the generation of the final feature map ($F_{\text{final}}$). $F_{\text{final}}$ serves as the input for the dual-branch detection head. Each branch of the detection head is composed of three CBS modules and one convolutional layer, enabling the extraction of discriminative features and subsequent prediction of ship centers and the encoded parameters. This design ensures that the detection head effectively captures the necessary information for accurate oriented ship detection in SAR images. The prediction process can be represented as follows:

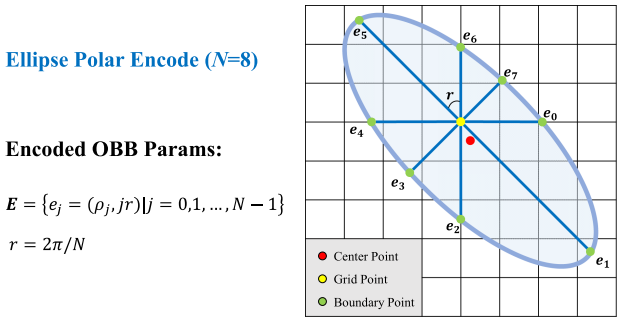



Fig. 2. Ellipse PE.

$$F_{\text{final}} = \text{CBS}_{1\times1}\ (F_{\text{neck}}) \tag{1}$$

$$H = \text{Conv}_{1\times1}\ (\text{CBS}_{1\times1}\ (\text{CBS}_{3\times3}\ (\text{CBS}_{3\times3}\ (F_{\text{final}})))) \tag{2}$$

$$E = \text{Conv}_{1\times1}\ (\text{CBS}_{1\times1}\ (\text{CBS}_{3\times3}\ (\text{CBS}_{3\times3}\ (F_{\text{final}})))) . \tag{3}$$

The final detection result is obtained by decoding the two prediction branches. The following will introduce the encoding and decoding methods of EPE in detail.

## B. Ellipse PE

*1) Encoding Process:* The ship has an ellipse-like shape, with the ellipse being a type of conic section that is smoother than a rotated rectangle. Hence, we propose utilizing an ellipse as the bounding box representation for ships, allowing for arbitrary positions and orientations. To address the boundary discontinuity problem, we encode the ellipse as a set of parameters for regression during network training. The encoding process is depicted in Fig. 2, and it involves the following specific steps.

First, we obtain the parameters of the target ground truth: $(\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}, \hat{\theta})$. We adopt the OpenCV convention, where $(\hat{c}_x, \hat{c}_y)$ represents the center point of the target, $\hat{w}$ and $\hat{h}$ represent the width and height of the target, and $\hat{\theta}$ represents the angle formed by the $X$-axis, whose range is $(0, \pi/2]$. We then transform these parameters to the scale of the feature map: $(c_x, c_y, w, h, \theta)$, where $c_x = \hat{c}_x/s$, $c_y = \hat{c}_y/s$, $w = \hat{w}/s$, $h = \hat{h}/s$, and $\theta = \hat{\theta}$. Here, $s$ represents the downsampling ratio, and in this article, $s = 8$.

By determining the grid point $(g_x, g_y)$ near the center point $(c_x, c_y)$, we establish a local 2-D coordinate system with it as the origin. We then convert the bounding box parameters to ellipse parameters, resulting in the elliptical equation

$$\frac{((x-o_x)\cos\theta+(y-o_y)\sin\theta)^2}{a^2}+\frac{((x-o_x)\sin\theta-(y-o_y)\cos\theta)^2}{b^2}=1 \tag{4}$$

where $(x, y)$ represents the coordinates of points on the ellipse. $\theta$ refers to the angle value of clockwise rotation of the ellipse around the center point, whose range is (0, $\pi$/2]. $o_x = c_x - g_x$ and $o_y = c_y - g_x$. $(o_x, o_y)$ represents the coordinate of the center of the ellipse in the local 2-D coordinate system, which indicates the position of the ellipse center relative to the grid point. $a = w/2$ and $b = h/2$. $a$ and $b$ represent the semimajor axis length and semiminor axis length of the ellipse, respectively.

We represent Cartesian coordinates $(x, y)$ using polar coordinates $(\rho, r)$. The equation of the ellipse can then be expressed in the following form:

$$\frac{((\rho\cos r-o_x)\cos\theta+(\rho\sin r-o_y)\sin\theta)^2}{a^2}+\frac{((\rho\cos r-o_x)\sin\theta-(\rho\sin r-o_y)\cos\theta)^2}{b^2}=1. \tag{5}$$

When the polar coordinate angle value $r$ is given, this equation becomes a quadratic equation in terms of the polar coordinate length value $\rho$

$$A\rho^2+B\rho+C=0$$

$$\begin{cases} A=a^2(\cos r\sin\theta-\sin r\cos\theta)^2 \\ \quad +b^2(\cos r\cos\theta+\sin r\sin\theta)^2 \\ B=2a^2(\cos r\sin\theta-\sin r\cos\theta)(o_y\cos\theta-o_x\sin\theta) \\ \quad -2b^2(\cos r\cos\theta+\sin r\sin\theta)(o_x\cos\theta+o_y\sin\theta) \\ C=a^2(o_y\cos\theta-o_x\sin\theta)^2 \\ \quad +b^2(o_x\cos\theta+o_y\sin\theta)^2-a^2b^2. \end{cases} \tag{6}$$

By solving the quadratic equation, we can obtain the distance from $(g_x, g_y)$ to any point on the ellipse

$$\rho=f(r)=\frac{-B+\sqrt{B^2-4AC}}{2A}. \tag{7}$$

By uniformly sampling $N$ values of $r$ within the interval [0, $2\pi$) and using $f(r)$, we can calculate $N$ corresponding values of $\rho$. This will result in a set of points on the ellipse

$$\boldsymbol{e}=\{(\rho_i, r_i)\,|\rho_i=f(r_i),\ r_i=2i\pi/N, i=0,1,2,\ldots,N-1\}. \tag{8}$$

In our method, it is important to note that a consistent set of polar coordinate angle values is employed across all OBBs. This implies that the total number of angles $N$ remains constant. Consequently, each individual OBB can be uniquely characterized by a distinct set of polar coordinate length values. This approach ensures that all OBBs share a common angular representation, allowing for a standardized and systematic analysis of their geometrical properties.

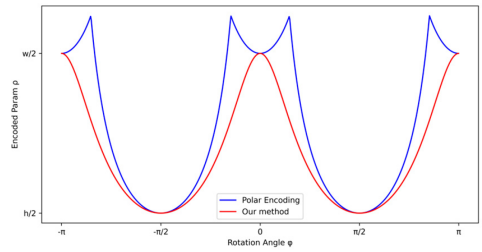



Fig. 3. Variation of encoded parameters.

*2) Influence of the Hyperparameter* N*:* In a plane, five points determine a conic [39]. Therefore, the parameter $N$ must be greater than or equal to 5 in order to satisfy the decoding condition. This ensures that an adequate number of points is available for the precise reconstruction of the OBB.

Furthermore, the value of $N$ directly affects the angular sampling rate, which plays a crucial role in the encoding process of the OBB. Insufficient sampling caused by an excessively low sampling rate may result in an incomplete representation of the ship's shape. Conversely, an excessively high sampling rate would reduce computational efficiency and increase the complexity of training. The selection of an appropriate angular sampling rate is vital to accurately capture the shape and unique features of the ship.

In Section V, we will offer further insights into the determination of an appropriate $N$, striking a balance between computational efficiency and the accurate portrayal of the ship's shape.

*3) Boundary Continuity of EPE:* Fig. 3 illustrates the variations of our EPE encoded parameters (red line) and the PE encoded parameters (blue line) with respect to the rotation angle $\varphi$ [8]. From the graph, several important observations can be made: First, the encoded parameters exhibit a periodic function with a period of $\pi$, ensuring continuity throughout the range of rotation angles. Second, the extreme values of our EPE parameters directly correspond to the width and height information of the OBB. Furthermore, a comparison between the variations of our EPE method and the PE method can be observed: First, our proposed method demonstrates continuous differentiability, meaning that the parameter variation is smooth and differentiable at all points, which is not the case for the PE method. Second, when the rotation angle $\varphi$ lies in the range (0, $\pi$/2], our method ensures a unique mapping between the parameters and angle values, providing a distinct and unambiguous representation. In contrast, the PE method does not possess this property. These findings underscore the advantages of our EPE method in terms of accurate representation, unique parameter mapping, and smooth convergence of the ship's bounding box.

The variation of the rotation angle, which means the angle error, causes a change in the encoding parameters. We represent this change as $S$ [8]. The calculation is given as follows:

$$S(\varphi)=\sum_{i=1}^{N}|\rho_i(\varphi)-\hat{\rho}_i| \tag{9}$$

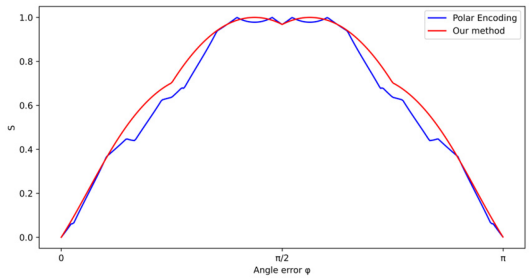



Fig. 4. Curves of $S(\varphi)$ with the aspect ratio of 2.

where $\varphi$ denotes the angle error, $\rho_i(\varphi)$ represents each encoded parameter, and $S(\varphi)$ represents the sum of the absolute errors of the encoded parameters.

Fig. 4 illustrates the curves of $S(\varphi)$ generated by our EPE method (red line) and the PE method (blue line) in the context of representing a detection box with the aspect ratio of 2 using 16 boundary points. The function depicted in the figure represents the $L1$ loss, enabling an examination of the relationship between angle errors and the resulting loss in the encoded parameters. To facilitate analysis, we normalize the values of $S$ within the range of [0, 1]. Our observations are as follows: First, in the interval [0, $\pi$], the loss initially increases from 0 and, subsequently, decreases back to 0. This behavior aligns with expectations, as the overlap between OBBs is minimized when the angle error is $\pi/2$, while it is maximized when the angle error approaches 0 or $\pi$. Second, notably, our proposed method exhibits smoother variations in loss compared with the PE method, indicating an advantage in terms of convergence.

In summary, for oriented ship detection, our method can generate a unique group of continuous encoded parameters and achieve reasonable and smooth loss regression, thereby overcoming the boundary discontinuity problem.

### C. Ellipse Polar Decoding

*1) Decoding Process:* Ellipse polar decoding is the inverse process of encoding. It involves extracting the center of ship and decoding the corresponding EPE parameters to obtain the ship detection box.

The decoding process during both the training and inference stages is essentially the same, with the only difference being the method of obtaining the centers of ships. During training, all the ship centers are known during the positive sample generation phase. During inference, locations in the center heatmap output $H$ that satisfy a score threshold represent ship centers. The detected $K$ ship centers can be represented as follows:

$$\boldsymbol{ctr} = \{(cx_i,\ cy_i)\,|i = 0, 1, \ldots, K-1\}. \tag{10}$$

To extract the $N$-channel values corresponding to the encoded parameters of each ship target in the EPE parameters map $E$, denoted as $\boldsymbol{\rho_i} = (\rho_0, \rho_1, \ldots, \rho_{N-1})_i$, we can represent the set of boundary points of the ellipse for the $i$th ship target in polar coordinates, with its center points as the origin. Specifically, it

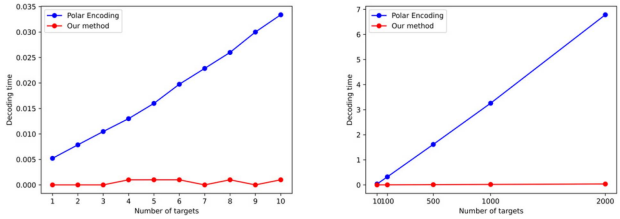



Fig. 5. Decoding time with different numbers of targets.

can be represented as follows:

$$(\boldsymbol{\rho_i}, \boldsymbol{r}) = \{(\rho_j, jr)\ |\ j = 0, 1, \ldots, N-1\} \tag{11}$$

where $r = 2\pi/N$ represents the sampling angular interval.

Next, by converting the polar coordinates to Cartesian coordinates and superimposing them on the corresponding centers, we obtain the set of elliptical boundary points for each ship target in the image

$$\begin{cases} \boldsymbol{x_i} = cx_i + \boldsymbol{\rho_i} \cos \boldsymbol{r} \\ \boldsymbol{y_i} = cy_i + \boldsymbol{\rho_i} \sin \boldsymbol{r}. \end{cases} \tag{12}$$

By using the set of ellipse boundary points $\boldsymbol{ebps_i} = (\boldsymbol{x_i}, \boldsymbol{y_i})$ for each ship target, we can fit the corresponding ellipse

$$\boldsymbol{ellipse_i} = \text{fitEllipse}\,(\boldsymbol{epts_i}). \tag{13}$$

The fitted ellipse $\boldsymbol{ellipse_i} = (x, y, w, h, \theta)$ can be used to represent the detection bounding box of the ship target.

*2) Decoding Efficiency:* To evaluate the effectiveness of our method during inference, we conduct decoding speed experiments comparing it with the PE method as a baseline algorithm. To ensure a fair comparison, both methods are tested on the same experimental platform using identical ground truths for encoding. The decoding time for each method is carefully measured and recorded for different numbers of targets, as depicted in Fig. 5. Our analysis reveals several significant observations. First, our method outperforms the PE method in terms of decoding time for a single ship target. This indicates a clear advantage of our approach in terms of speed and efficiency. Furthermore, the decoding time of the PE method increases as the number of targets grows, resulting in a longer inference time. Conversely, our method exhibits minimal sensitivity to the number of targets, with little impact on the decoding time. Overall, these findings provide strong empirical evidence supporting the real-time inference capabilities of our method.

### D. Heatmap and NMS

*1) Heatmap:* CenterNet requires anticipating the position of the target's center point, which involves generating a target center heatmap using a standard 2-D Gaussian distribution centered on the corresponding position of the target's center point [27]. The heatmap value increases as the predicted point gets closer to the target center. The center heatmap branch calculates the loss by comparing the prediction with the target, guiding the network's training process. However, this approach barely considers the scale information of the target, neglecting its shape and direction, making it challenging to accurately represent ship

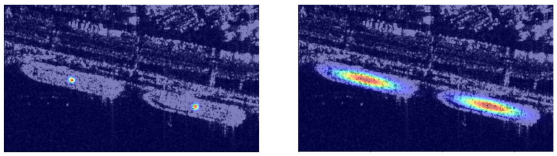



Fig. 6. Heatmap. (a) Standard 2-D Gaussian. (b) Ellipse 2-D Gaussian.

targets with large aspect ratios and arbitrary orientations in SAR images.

To address this limitation and incorporate the ship's orientation and shape information into the target heatmap, we employ an ellipse 2-D Gaussian distribution. This modification enables the network to learn the distribution of features related to the ship's shape and direction, enhancing its ability to capture ships with diverse aspect ratios and orientations in SAR images.

With the ground truth at feature map scale $(c_x, c_y, w, h, \theta)$, the corresponding covariance matrix can be calculated as follows:

$$\boldsymbol{\Sigma} = \boldsymbol{R}\boldsymbol{\Lambda}^2\boldsymbol{R}^{\mathrm{T}} \tag{14}$$

$$\boldsymbol{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{15}$$

$$\boldsymbol{\Lambda} = \begin{pmatrix} w/2 & 0 \\ 0 & h/2 \end{pmatrix}. \tag{16}$$

Thus, we can obtain the 2-D ellipse Gaussian heatmap $\boldsymbol{G}$ corresponding to each ship in the SAR image

$$\boldsymbol{G} = \exp\left(-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{u})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{u})\right)^{\mathrm{T}} \tag{17}$$

where $\boldsymbol{X} = (x, y)$ represent the positions of all points on the feature map. $\boldsymbol{u} = (g_x, g_y)$ represents the grid point near the center point of ship $(c_x, c_y)$. $\boldsymbol{\Sigma}^{-1}$ represents the inverse of covariance matrix. The ellipse Gaussian heatmaps of all ships in the same SAR image are combined, and the maximum value is taken for each position to obtain the center heatmap $\hat{H}$ of this image, as illustrated in Fig. 6.

*2) Nonmaximum Suppression:* NMS plays a pivotal role in object detection [36]. NMS is essential for refining object detection results by eliminating duplicate detections and selecting the most confident bounding boxes. This process enhances the accuracy of ship detection and reduces false positives.

### E. Loss Function

The training loss is divided into two parts, corresponding to the two branches of oriented ship detection.

For the center prediction branch $H$, we adopt the same loss function as CenterNet [27]. The loss calculation formula is given as follows:

$$L_H = -\frac{1}{K}\sum_{xy}\begin{cases}(1-h_{xy})^{\alpha}\log h_{xy} \; if \;\; \hat{h}_{xy}=1 \\ \left(1-\hat{h}_{xy}\right)^{\beta} h_{xy}^{\alpha}\log\left(1-h_{xy}\right) \;\; \text{otherwise}\end{cases} \tag{18}$$

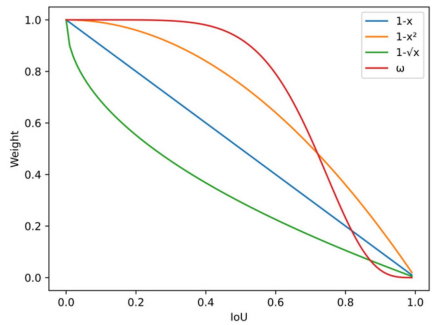



Fig. 7. Different IOU weighting functions.

where $h_{xy}$ represents the predicted value of the center heatmap $H$ at position $(x, y)$; $\hat{h}_{xy}$ represents the value of the label heatmap $\hat{H}$ at position $(x, y)$; $\alpha$ and $\beta$ are the hyperparameters controlling the easy and hard sample balancing, which are, respectively, set to 2 and 4; and $K$ is the number of ship targets.

Gao et al. [35] use smooth $L1$ loss to calculate the loss of the encoding branches, but in most cases, the SAR ship is not a perfect elliptical shape. Therefore, we employ an IOU function to balance the regression of encoded parameters, as shown in (15). The curve of this function is illustrated by the red line in Fig. 7. The meaning of this curve can be explained as follows: Ships exhibit arbitrary orientations and large aspect ratios, resulting in significant changes in IOU with even small variations in angles. Hence, when the IOU is not sufficiently large, it is necessary to focus on the loss of encoding parameters to ensure network convergence. However, when the IOU is large enough, SAR image targets often exhibit characteristics, such as speckle and deformation, and the ship bounding box is not a true ellipse. In such cases, the encoding parameters may not accurately represent the ground truth values, requiring the suppression of the encoding parameter loss to prevent overfitting

$$\omega = 0.25\left(\cos\left(\mathrm{IOU}^3\pi\right)+1\right)^2. \tag{19}$$

At the same time, we utilize label heatmap weighting to balance the positive samples of the same ship target at different positions. The calculation formula is given as follows:

$$L_E = \frac{1}{M}\sum_{i=1}^{M}\hat{h}_i^2\omega_i \mathrm{Smooth}_{L1}\left(\boldsymbol{\rho_i},\ \hat{\boldsymbol{\rho}}_{\boldsymbol{i}}\right)) \tag{20}$$

where $M$ is the number of positive samples, and $\boldsymbol{\rho_i}$ and $\hat{\boldsymbol{\rho}}_{\boldsymbol{i}}$ represent the predicted and ground truth encoding parameters of the $i$th ship target ellipse, respectively. $\omega_i$ and $\hat{h}_i$ denote the loss weight and label heatmap value of the $i$th ship.

Finally, the two parts of the loss functions are summed up to obtain the final loss function

$$\mathrm{Loss} = L_H + L_E. \tag{21}$$

## IV. Experiment

In this section, we provide a comprehensive report on the experiments conducted using open-source datasets to validate

the effectiveness of our method. First, we provide an overview of the dataset, illustrate the experimental setup, and introduce the evaluation metrics employed in this study. Then, we corroborate our network selection through backbone network comparison experiments and validate the efficacy of our detection method through ablation experiments. Finally, we present comparative experiments pitting our method against other oriented target detection methods, analyze the experimental results, and evaluate the overall effectiveness of our approach.

### A. Dataset Description

*1) RSDD-SAR Dataset:* The RSDD-SAR dataset [20], which is publicly accessible for oriented SAR ship detection, involves data obtained from the domestic Gaofen-3 satellite and TerraSAR-X satellite. It comprises a collection of 7000 SAR images, encompassing diverse imaging modes, polarization modes, and resolutions. Within this dataset, a total of 10263 ship instances are annotated. The rotation boxes in the dataset are annotated using the long-edge definition method, ensuring consistent and accurate representation. The ship target angles exhibit a uniform distribution ranging from $-90°$ to $90°$, reflecting the arbitrary orientations of the ships. Moreover, the aspect ratios primarily range between 1.5 and 7.5, emphasizing the prevalence of large aspect ratios. For comprehensive information regarding the dataset, we refer readers to [20]. The dataset is partitioned into a default training set consisting of 5000 images and a test set comprising 2000 images. The test set is further subdivided into 159 images for inshore testing and 1841 images for offshore testing, enabling a thorough evaluation of the proposed method's performance.

*2) SSDD Dataset:* The SSDD dataset is composed of 1160 SAR images, featuring various resolutions, polarization modes, and sea surface conditions [40]. Within this dataset, there are a total of 2540 SAR ship targets, with an average of approximately 2.12 ship targets per SAR image. The inclusion of multiple resolutions ensures that the training model can effectively detect multiscale targets. The SSDD dataset encompasses a range of scenes, including those with more land clutter in inshore scenarios and less clutter in offshore ones. Detecting ship targets in inshore scenes can be more challenging due to the increased land clutter, and this subset of images is smaller in number. The dataset is divided into two main parts: a default training set, consisting of 928 images, and a test set, which comprises 232 images. To assess the model's detection performance across different scenes, the test set is further categorized into two subsets: inshore (39 SAR images) and offshore (193 SAR images).

### B. Experimental Settings

*1) Platform Configuration:* The experiments are conducted on an Ubuntu 20.04 operating system, utilizing hardware comprising an Intel Xeon Platinum 8255C CPU and an NVIDIA GeForce RTX 2080Ti GPU. All the compared algorithms for SAR image rotation box detection are implemented using the JDet [41] framework and MMRotate [42] framework. To ensure a fair and consistent comparison, data augmentation techniques only involve horizontal random flipping with a probability of 0.5. The optimizer utilized in this study is SGD [43], with the initial learning rate set to 0.0025, a momentum of 0.9, and a weight decay of 0.0001. To prevent exploding gradients, $L2$-norm gradient clipping is employed, and the maximum gradient norm is capped at 35. The iteration rounds are configured at 12, with the learning rate undergoing a tenfold reduction in the seventh and tenth iteration rounds. In addition, a linear warm-up learning rate strategy is implemented for the first 500 batches, where the initial warm-up learning rate is set to one-third of the initial learning rate. Detailed parameter configurations adhere to the settings specified in [20].

TABLE I
COMPARATIVE EXPERIMENTS OF BACKBONES

| Backbone | Params(M) | $AP_{50}$(%) |
|---|---|---|
| HRnet | 30.15 | 79.77 |
| Res2Net-101 | 57.55 | 88.41 |
| Darknet-53 | 46.35 | 78.51 |
| CSPDarknet-53 | 30.69 | 86.03 |
| ResNet-101 | 46.35 | 88.95 |
| ResNet-50 | 27.35 | 89.17 |
| ResNet-18 | 11.92 | 87.95 |
| ResNet-50(no C5) | **11.86** | **89.37** |

The bolded values represent the optimal values within each column.

*2) Evaluation Metrics:* The evaluation metrics for the model include the model parameters size, AP50 for inshore scenes, AP50 for offshore scenes, and overall AP50. AP50 represents the average precision (AP) at an IOU threshold of 0.5. AP50 is defined as follows:

$$\begin{cases} P = \frac{N_{td}}{N_d} \\ R = \frac{N_{td}}{N_r} \\ \mathrm{AP} = \int_0^1 P(R)\, dR \end{cases} \tag{22}$$

where $N_{td}$ represents the number of the correctly detected targets, $N_d$ represents the total number of the detected targets, and $N_r$ represents the actual number of the targets; $P$ represents the precision, $R$ represents the recall, and AP represents the area under the $P - R$ curve.

The practical effectiveness of modes depends on just as much on their inference speed. Within this article, we estimate the model's efficiency using two pivotal metrics: frames per second (FPS) and floating-point operations per second (FLOPs). FPS denotes the number of images the detection network can process in 1 s, with higher value indicating faster detection speed. FLOPs are employed as an indicator of the model's time complexity, enabling us to compare processing times across different hardware platforms.

### C. Backbone Experiments

We test the experimental results of our method under different backbones, as shown in Table I. The backbones encompass HRnet [44], Res2Net [45], Darknet53 in YOLOv3 [24], CSPDarknet in YOLOX [46], and the ResNet series [37]. Our EPE method avoids predicting the rotation angle and encodes OBB

TABLE II
ABLATION EXPERIMENTS OF OUR METHOD ON RSDD-SAR

| Model | $AP_{50}$(%) | Inshore $AP_{50}$(%) | Params(M) |
|---|---|---|---|
| baseline | 87.21 | 56.53 | 14.04 |
| +EPE Head | 87.84(+0.63) | 62.62(+6.09) | 11.86 |
| +Ellipse Gaussian Heatmap | 89.05(+1.21) | 65.60(+2.98) | 11.86 |
| +EPE Loss | 89.37(+0.32) | 67.84(+2.24) | 11.86 |

into EPE parameters with intuitive geometric meaning, enabling the network to learn prediction. This eliminates the need for an excessively deep feature extraction backbone. As can be seen from the table, our choice, "ResNet50(no C5)," achieves the optimal performance in terms of parameters and accuracy.

### D. Ablation Experiments

To verify the effectiveness of our method, we conduct ablation experiments on the RSDD dataset. The experimental results are shown in Table II. We replace the R-FCOS head into our network model as our baseline. The backbone and neck of our baseline are ResNet50 (no C5) and FPN. On this baseline, we gradually add EPE head, ellipse Gaussian heatmap, and EPE loss to verify the effectiveness of our method.

*1) EPE Head:* We replace the FCOS head with our EPE head, the detection mechanism is our EPE method, the heatmap generation method is the standard 2-D Gaussian, and the loss function of the encoding parameter branch is smooth $L1$. It can be concluded from Table II that the EPE method has brought significant improvements to our model. The AP50 in the test dataset and the inshore dataset increases by 0.63% and 6.09%, respectively. And the model parameters were reduced by 2.18M. This shows that our method can effectively improve the accuracy of inshore ship detection.

*2) Ellipse Gaussian Heatmap:* Based on the aforementioned setting, we use the ellipse Gaussian heatmap as the label of the center prediction branch, and the detection accuracy is improved. It is proved that the ellipse Gaussian heatmap enables the model to effectively obtain the overall feature distribution of ships and improve ship detection performance.

*3) EPE Loss:* We add EPE loss to avoid overfitting of the predictive encoding parameters. This produces a 2.24% AP50 improvement for the inshore dataset.

### E. Comparative Experiments

To evaluate the effectiveness of our proposed EPE method, this article conducts a comparative analysis with the state-of-the-art oriented target detection methods, which can be categorized into three main groups.

1) Anchor-based methods, including R-faster R-CNN [9], ROI transformer [10], oriented R-CNN [11], ReDet [12], S2ANet [13], R3Det [14], and R-RetinaNet based on RetinaNet [23].
2) Angle-based methods, represented by R-FCOS based on FCOS [26].
3) Encoding-based methods, encompassing gliding vertex [18], CFA [29], and PE [8].

By comparing our proposed method with these diverse approaches, we aim to provide a comprehensive evaluation of its performance and highlight its advantages over the existing methodologies.

The experimental results for all comparative algorithms, including our proposed method, are presented in Table III and Figs. 8 and 9. All comparative algorithm experimental results originate from the RSDD-SAR article [20]. It can be observed that, among these comparative algorithms, the anchor-based methods generally have better performance in offshore detection accuracy, while the results for inshore detection vary. The angle prediction-based methods, due to the issue of boundary discontinuity, perform relatively poorly in nearshore scenarios. The coding-based CFA method achieves the highest inshore detection accuracy among all the compared algorithms. However, our proposed method surpasses these approaches in terms of both parameter efficiency and nearshore detection performance. This suggests that our method achieves superior performance in oriented ship detection without necessitating complex and intricate network architectures.

The one-stage target detection algorithm outperforms the two-stage algorithm in terms of efficiency primarily because it condenses the two stages of the latter into a single stage. Table IV provides insights into the inference speed of several one-stage algorithms, including our own method. Notably, due to the simplicity of our model, it exhibits the lowest FLOPs. However, during decoding, there is a demand to fit ellipses, which means that while FPS may not be the highest, it remains capable of meeting real-time monitoring requirements. Furthermore, our method significantly outpaces the PE method in terms of inference speed.

Fig. 8 illustrates the detection outcomes of different methods in the offshore region. The results clearly demonstrate our method's proficiency in effectively detecting all types of ships, including numerous small-sized vessels, while precisely determining their orientations and dimensions. This underscores the robustness and efficacy of our approach in detecting oriented ship targets in offshore scenarios, even in the presence of challenging conditions, such as a high concentration of small ships.

In the nearshore area, the complexities of ship detection are significantly heightened. It is in these demanding conditions that the effectiveness of our detection method can be most thoroughly evaluated. Fig. 9 indicates the experimental results from various algorithms in the inshore region, providing valuable insights into their performance. Overall, our method exhibits superior detection capabilities compared with all other comparative algorithms, with only the CFA method showing a slightly lower performance. To further dissect the detection performance of our

TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON RSDD-SAR

| Categories | Model | Backbone | Params(M) | Inshore $AP_{50}$(%) | Offshore $AP_{50}$(%) | $AP_{50}$(%) |
|---|---|---|---|---|---|---|
| Anchor-based | R-Faster R-CNN | ResNet-50 | 41.41 | 48.78 | 90.93 | 83.29 |
| | ROI Transformer | ResNet-50 | 55.32 | 60.83 | **94.35** | 88.39 |
| | Oriented R-CNN | ResNet-50 | 41.35 | 65.92 | 90.21 | 88.84 |
| | ReDet | ResNet-50 | 31.57 | 61.94 | 90.34 | 88.40 |
| | S2ANet | ResNet-50 | 36.45 | 63.27 | 93.14 | 87.91 |
| | R3Det | ResNet-50 | 41.81 | 56.87 | 90.16 | 80.87 |
| | R-RetinaNet | ResNet-50 | 32.44 | 33.20 | 74.06 | 66.66 |
| Angle-based | R-FCOS | ResNet-50 | 32.17 | 50.02 | 93.09 | 85.48 |
| Encoding-based | Gliding Vertex | ResNet-50 | 41.41 | 55.93 | 91.65 | 85.55 |
| | CFA | ResNet-50 | 36.83 | 66.40 | 90.47 | 89.31 |
| | Polar Encoding | ResNet-50 | 52.83 | 59.69 | 90.12 | 87.31 |
| | Ours | ResNet-50 | 27.35 | 67.25 | 90.31 | 89.17 |
| | Ours | ResNet-50(no C5) | **11.86** | **67.84** | 90.37 | **89.37** |

The bolded values represent the optimal values within each column.

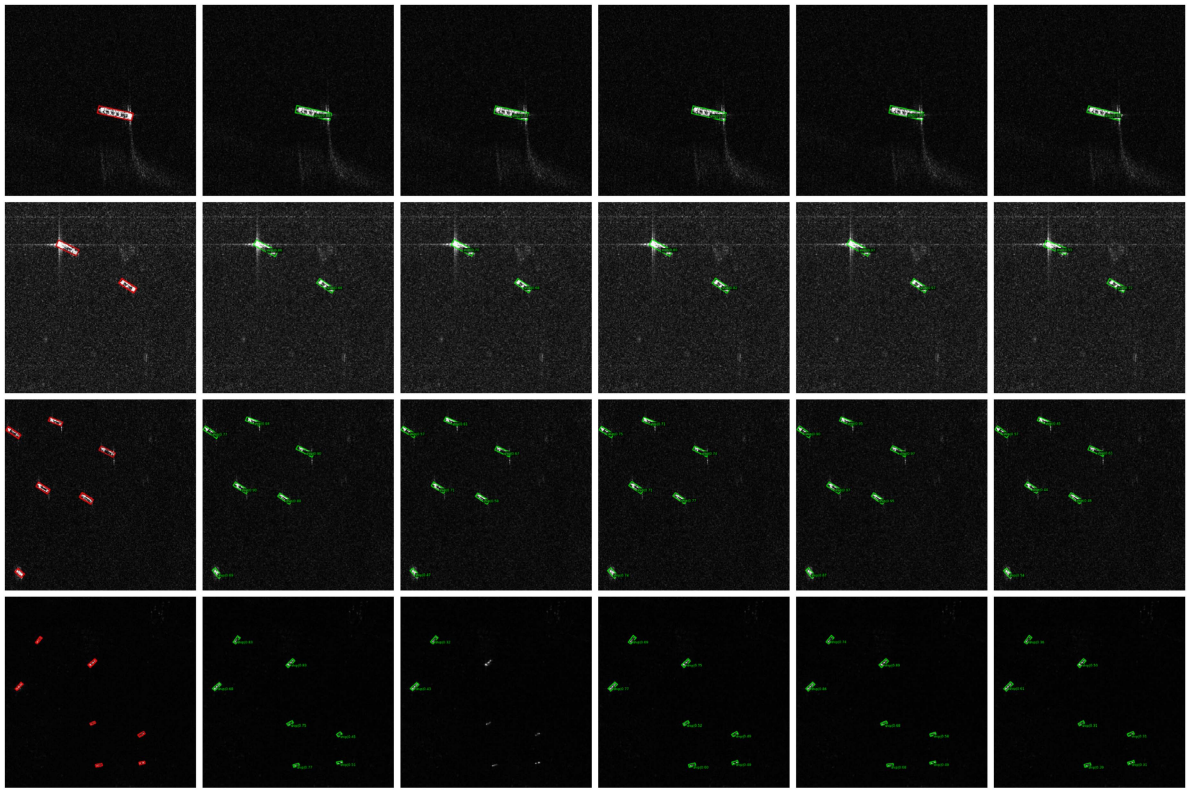



Fig. 8. Detection results of different methods in the offshore region on the RSDD-SAR dataset. (a) Ground truth. (b) Our method. (c) CFA. (d) R-FCOS. (e) R3Det. (f) R-RetinaNet.

method in the nearshore area, we employ the CFA method as a baseline for comparison.

During the detection process in the inshore region, the CFA method may overlook certain ships, especially those with less prominent features or those situated close to the shoreline. In contrast, our method excels in detecting such ships and accurately estimating their orientation, width, and height. These results underscore the robustness and effectiveness of our approach in detecting ships within the challenging nearshore environment, particularly for ships with less distinct characteristics and those located near the coast.

In the harbor region, oriented ship detection methods encounter challenges in distinguishing between buildings and docked vessels, as well as in effectively discerning adjacent ships. The CFA method, in particular, struggles with this differentiation, leading to instances of missed detections and false

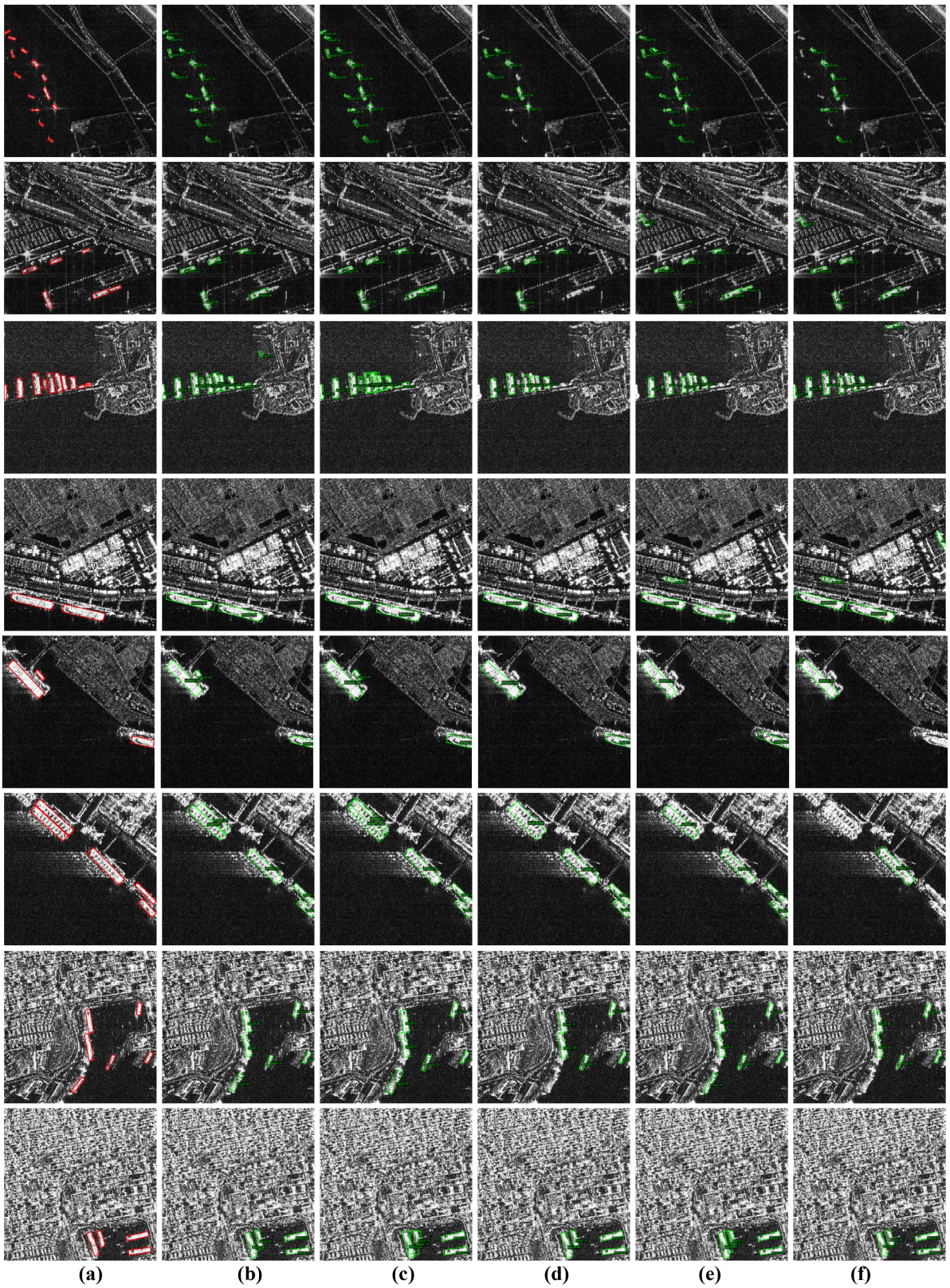



Fig. 9. Detection results of different methods in the inshore region on the RSDD-SAR dataset. (a) Ground truth. (b) Our method. (c) CFA. (d) R-FCOS. (e) R3Det. (f) R-RetinaNet.

TABLE IV
INFERENCE SPEED ON RSDD-SAR

| Method | Params(M) | Flops(G) | FPS | $AP_{50}$(%) |
|---|---|---|---|---|
| R-FCOS | 32.17 | 51.55 | 44.2 | 85.48 |
| R-RetinaNet | 32.44 | 52.39 | 39.9 | 66.66 |
| CFA | 36.83 | 48.56 | 36.5 | 89.31 |
| R3Det | 41.81 | 82.17 | **46.1** | 80.87 |
| Polar Encoding | 52.83 | 115.37 | 18.7 | 87.31 |
| Ours | **11.86** | **31.0** | 39.3 | **89.37** |

The bolded values represent the optimal values within each column.

TABLE V
COMPARATIVE EXPERIMENTS ON SSDD

| Method | Params(M) | FLOPs(G) | FPS | $AP_{50}$(%) |
|---|---|---|---|---|
| R-FCOS | 31.89 | 51.55 | 44.2 | 47.82 |
| R3Det | 41.58 | 82.17 | 46.1 | 46.10 |
| R-Retinanet | 36.13 | 52.39 | 39.9 | 27.29 |
| CFA | 32.6 | 48.56 | 36.5 | 64.95 |
| **Ours** | **11.87** | **31.01** | 38.1 | **72.24** |

The bolded values represent the optimal values within each column.

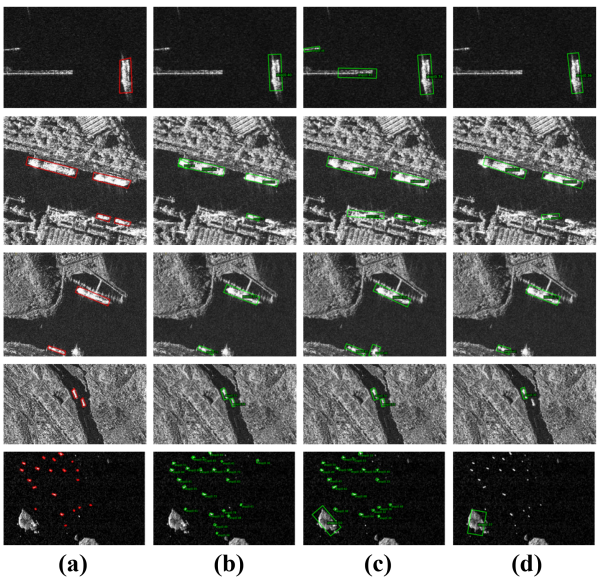



Fig. 10. Detection results of different methods on SSDD dataset. (a) Ground truth. (b) Our method. (c) CFA. (d) R-FCOS.

positives. Conversely, our proposed method excels in accurately detecting ships within the harbor area and provides precise information about their orientation, length, and width. These results emphasize the robustness and efficacy of our approach in detecting ships within complex harbor environments, effectively overcoming challenges related to object differentiation and accurate ship localization in densely packed surroundings.

Furthermore, we conduct experiments to assess the performance of our method on the SSDD dataset [40]. The experimental results of our method and the comparison algorithm on the nearshore test set are presented in Table V and Fig. 10. In contrast to the RSDD-SAR dataset, the SSDD dataset is smaller and contains fewer instances of ships. To make the most of the

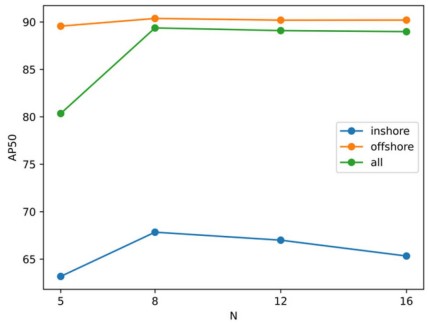



Fig. 11. Effect of $N$ on detection performance.

available features for ship targets, we modify the hyperparameter $N$ to 16. As observed in the table, our method attains the highest accuracy in nearshore detection while employing the fewest model parameters. This further underscores the outstanding performance of our approach in complex nearshore environments.

## V. DISCUSSION

We conduct tests to assess the detection performance of our method under different hyperparameters $N$ using the RSDD-SAR dataset, as shown in Fig. 11. The following observations can be made: First, in the offshore area, all $N$ values achieve high precision, with $N = 8$ being relatively higher. Second, in the inshore area, there is a noticeable performance difference among different $N$ values. $N = 5$ performs the worst, while $N = 12$ and $N = 16$ slightly underperform compared with $N = 8$. This discrepancy arises because the sampling rate of $N = 5$ is too low to fully capture the shape information of ships and guide network training, resulting in poorer performance in complex nearshore environments. When $N = 8$, the distribution of sampling points in polar coordinate encoding aligns well with the convolutional shape, leading to better detection performance. When $N = 12$ and $N = 16$, both detection performance and computational efficiency are inferior to $N = 8$. In conclusion, we recommend using $N = 8$ as the optimal choice.

The exceptional detection performance achieved by our proposed method can be attributed to two key factors. First, we devise the EPE specifically tailored for ship detection. This design choice yields a more distinct and direct network representation, reducing the number of prediction branches. This effectively minimizes the occurrence of coupling errors between coherent branches. Second, we devise the EPE loss specifically, which enhances the convergence speed during network training. Consequently, our method achieves remarkable detection performance without relying on complex and abstract network architectures, emphasizing the simplicity and effectiveness of our approach.

## VI. CONCLUSION

To address the boundary discontinuity problem in oriented SAR ship detection, we propose the EPE method. It treats the ship detection box as an ellipse and samples equiangular polar coordinate points, resulting in a smooth and uniquely mapped set of EPE parameters. Based on this approach, we design a lightweight DCNN for oriented SAR ship detection. During

training, the network undergoes end-to-end training and regression of ship centers and encoded parameters. In inference, the network efficiently decodes the predicted centers and encodes parameters to obtain ship detection results. We recommend using an IOU function weighted to calculate the encoded parameters loss, considering the speckle, deformation, and nonideal ellipse shape of the SAR ship. Experimental results on the RSDD-SAR dataset demonstrate the superiority of our method in terms of detection accuracy and parameter size, confirming its effectiveness. The EPE method's adaptability to ellipse-like targets makes it easily applicable to other visual tasks.

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